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$$
\begin{aligned}
&f(x) = \frac{1}{2(x-3)^4} + x \\
&f'(x) = -\frac{1}{(2(x-3)^4)^2} \cdot 8(x-3)^3 \cdot 1 + 1 \\
&= -\frac{1}{4(x-3)^8} \cdot 8(x-3)^3 + 1 \\
&= -\frac{8(x-3)^3}{4(x-3)^8} + 1 = -\frac{2(x-3)^3}{(x-3)^8} + 1 \\
&= -2(x-3)^{-5} + 1 = -\frac{2}{(x-3)^5} + 1
\end{aligned}
$$

References